

Postie Data Challenge

What the Transaction Data Shows — and What It Doesn't

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Presented by:

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Dataset Overview

This analysis uses three days of **checkout transactions** from **July 1–July 3, 2017**.

Each transaction includes a **timestamp**, **checkout amount**, **customer ID**, **website ID**, **app version**, and **checkout URL**.

The checkout URLs are not just page links — they appear to **encode product and quantity information**:

```
...?Round+Kumquat=1&Black%2FWhite+Pepper=1
```

This makes it possible to analyze the data at both the transaction level and the cart level.

Transaction level: daily sales, checkout values, website/app behavior

Cart level: products, quantities, cart size, combinations, inferred prices

1. Why was the 2017-07-03 sales value reported as much lower than the previous day?

The reported July 3 sales value appears **much** lower because it was calculated using the raw 2017-07-03.csv source file rather than UTC-normalized transaction dates.

Definition	July 3 Sales	Difference from Reported
Reported July 3 value	\$164,065	\$0
Raw source-file grouping	\$164,065	\$0
UTC-normalized transaction date	\$181,583	\$17,518

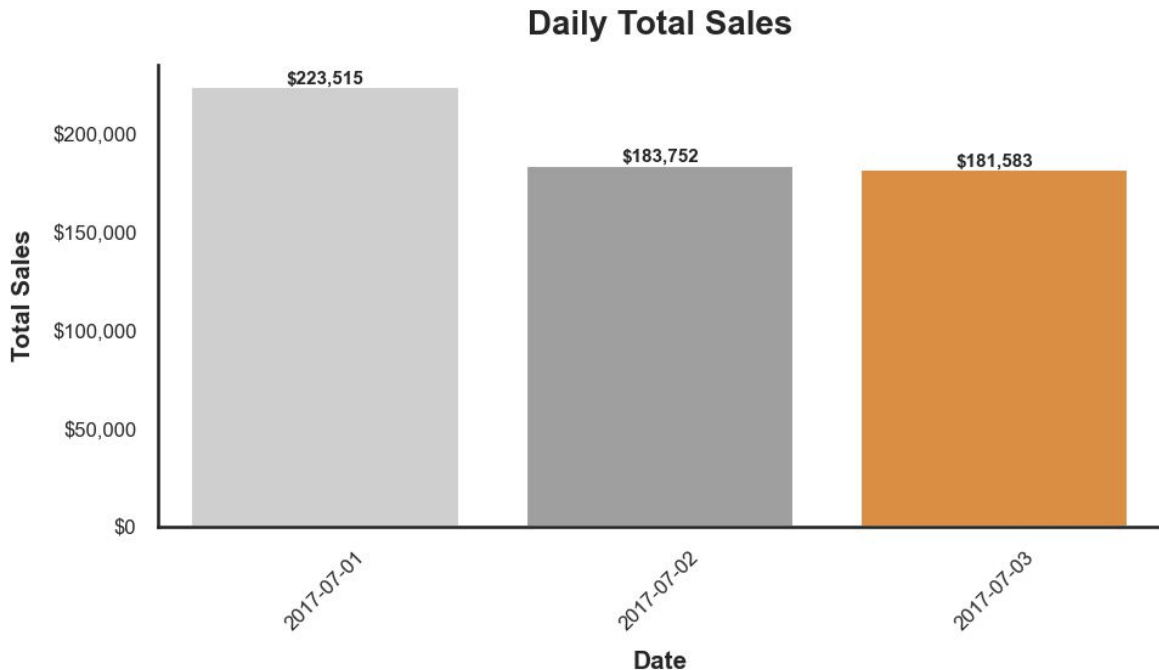
THE REPORTED VALUE MATCHES THE RAW SOURCE FILE — NOT THE NORMALIZED DATE TOTAL

1. Why was the 2017-07-03 sales value reported as much lower than the previous day?

Once transactions are grouped by UTC-normalized date, July 3 is still the lowest sales day,

but only slightly lower than July 2.

July 3 is \$2,169 lower than July 2, a decrease of about 1.2%



JULY 3 IS LOWER AFTER NORMALIZATION — BUT NOT BY MUCH

1. Why was the 2017-07-03 sales value reported as much lower than the previous day?

July 3 has a few unusual records worth flagging:

The single largest transaction occurs on day 3:

- This sale is for \$60,000 exactly
- The URL contains “error=True”

Timestamp	Website	App Version	Checkout Amount	Parsed Product	URL
2017-07-03 07:59:32 UTC	124	1.2	\$60,000	error	http://xyz.com/checkout?Bignay=1&error=True

For the following analysis, I have not removed this entry. We will revisit this transaction when discussing URL-derived cart data and system context.

JULY 3 HAS UNUSUAL RECORDS, BUT DATE HANDLING IS THE MAIN EXPLANATION

1. Why was the 2017-07-03 sales value reported as much lower than the previous day?

July 3 has a few unusual records worth flagging:

There are 17 transactions that are negative, only on day 3

- Each is exactly worth -\$12 (equalling -\$204 in total)
- These transactions occur across both app versions and both website IDs
- Parsed cart contents imply each negative transaction should total **+\$12**, suggesting the values may be sign-reversed rather than arbitrary errors.

Scenario	July 3 Sales
Keep -12 values as valid negative revenue	\$181,583
Exclude -12 values as errors	\$181,787
Treat -12 values as sign errors	\$181,991

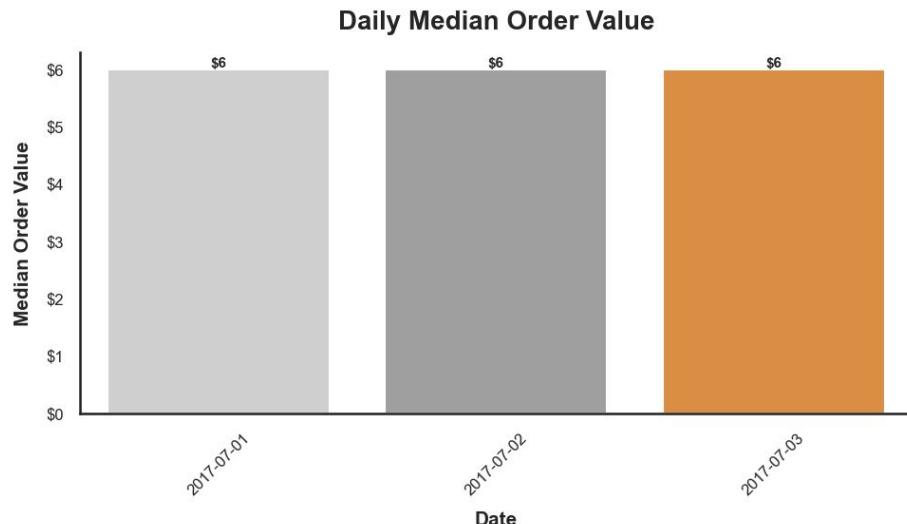
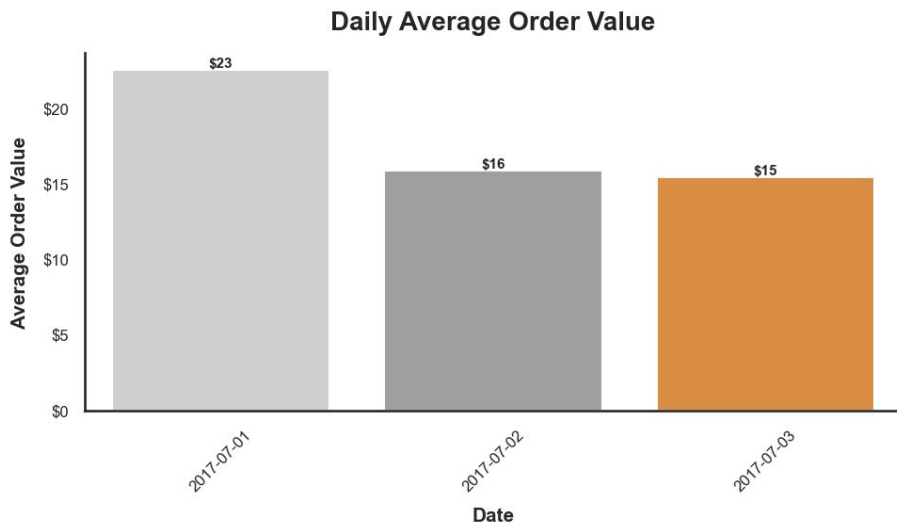
No matter how these values are handled, July 3 still reports the lowest total sales.

JULY 3 HAS UNUSUAL RECORDS, BUT DATE HANDLING IS THE MAIN EXPLANATION

2. What other metrics should the analyst report? Is average sales value the right metric?

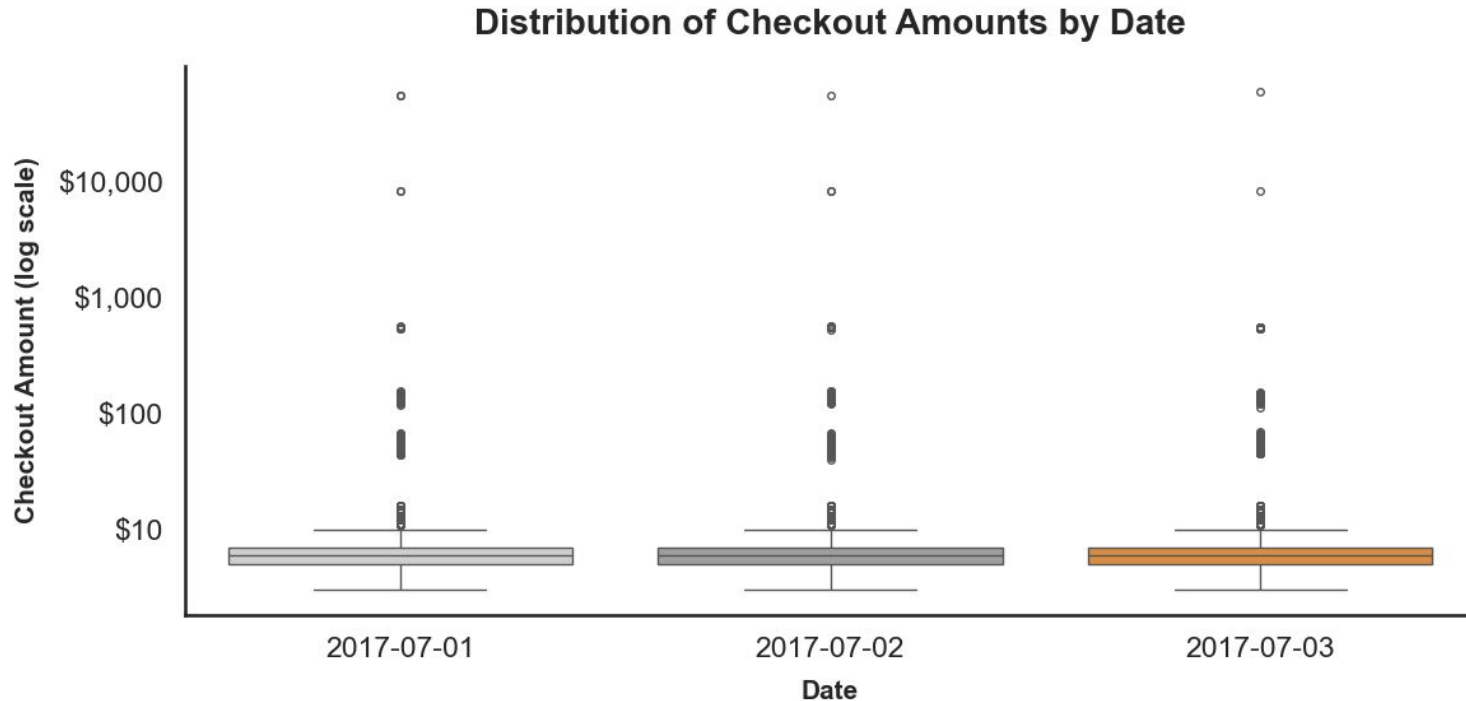
Average order value changes across days, but median order value stays fixed at \$6.

That suggests the average is being pulled by large transactions, not by a broad shift in typical checkout behavior.



AVERAGE ORDER VALUE IS USEFUL, BUT IT SHOULD NOT STAND ALONE

2. What other metrics should the analyst report? Is average sales value the right metric?



A FEW LARGE CHECKOUTS CAN MOVE THE AVERAGE MORE THAN THE TYPICAL ORDER

2. What other metrics should the analyst report? Is average sales value the right metric?



HOURLY SPIKES REINFORCE WHY TOTALS AND AVERAGES NEED CONTEXT

3. What information can be extracted from the URLs? Can product prices be inferred?

<http://xyz.com/checkout?Round+Kumquat=1&Black%2FWhite+Pepper=1>

URL pattern	Interpreted cart information
<code>...?Bignay=1</code>	1 Bignay
<code>...?Ume=1&Natal+Orange=1</code>	1 Ume + 1 Natal Orange
<code>...?Round+Kumquat=1&European+Grape=1</code>	1 Round Kumquat + 1 European Grape

The URLs contain **product** and **quantity** information in the query string.

This makes it possible to reconstruct cart contents from each transaction.

URLS TURN THE TRANSACTIONS INTO CART-LEVEL DATA

3. What information can be extracted from the URLs? Can product prices be inferred?

Product	Inferred Unit Price
Prairie Potato	\$3
Hazelnut	\$4
Black/White Pepper	\$5
European Grape	\$5
Ylang-ylang	\$5
Bignay	\$6
Natal Orange	\$6
Ume	\$6
Round Kumquat	\$7
Mabolo	\$8

Product Prices can be inferred

Single-product carts are the cleanest starting point for price inference because the checkout amount can be divided directly by the item quantity:

$$\text{checkout_amount} = \text{quantity} \times \text{unit_price}$$

For multi-product carts, the checkout amount is a combined total:

$$\text{checkout_amount} = \text{quantity_1} \times \text{price_1} + \text{quantity_2} \times \text{price_2} + \dots$$

URLS TURN THE TRANSACTIONS INTO CART-LEVEL DATA

4. Are there any interesting purchasing combinations, events, or metrics that are worth reporting and displaying?

	Product Combination	Transactions	Median Checkout	Average Checkout	Average Total Quantity	Days Observed
	Bignay + Black/White Pepper + European Grape + Hazelnut + Mabelo + Natal Orange + Prairie Potato + Round Kumquat + Ume + Ylang-ylang	161	\$140	\$1,485.99	270.0	3
	Prairie Potato + Ylang-ylang	71	\$8	\$8.00	2.0	3
	Black/White Pepper + Hazelnut	70	\$9	\$9.00	2.0	3
	Bignay + Black/White Pepper	70	\$11	\$11.00	2.0	3
	Hazelnut + Prairie Potato	67	\$7	\$7.00	2.0	3
	Bignay + Mabelo	66	\$14	\$14.00	2.0	3
	Hazelnut + Mabelo	65	\$12	\$11.63	2.0	3
	Natal Orange + Round Kumquat	64	\$13	\$13.00	2.0	3
	Mabelo + Ylang-ylang	64	\$13	\$13.00	2.0	3
	European Grape + Prairie Potato	64	\$8	\$8.00	2.0	3
	European Grape + Natal Orange	63	\$11	\$11.00	2.0	3
	Black/White Pepper + Prairie Potato	62	\$8	\$8.00	2.0	3
	Hazelnut + Ylang-ylang	61	\$9	\$9.00	2.0	3
	Hazelnut + Ume	61	\$10	\$10.00	2.0	3
	Prairie Potato + Round Kumquat	61	\$10	\$10.00	2.0	3

Most common multi-product cart:

Bignay + Black/White Pepper + European Grape + Hazelnut + Mabelo + Natal Orange + Prairie Potato + Round Kumquat + Ume + Ylang-ylang

Transactions: 161

Median checkout amount: \$140

Average checkout amount: \$1,485.99

Average total quantity: 270.0

Days observed: 3

THE MOST COMMON MULTI-PRODUCT CART CONTAINS ALL 10 PRODUCTS

4. Are there any interesting purchasing combinations, events, or metrics that are worth reporting and displaying?

Timestamp	Website	App Version	Checkout Amount	Parsed Product	URL
2017-07-03 07:59:32 UTC	124	1.2	\$60,000	error	http://xyz.com/checkout?Bignay=1&error=True

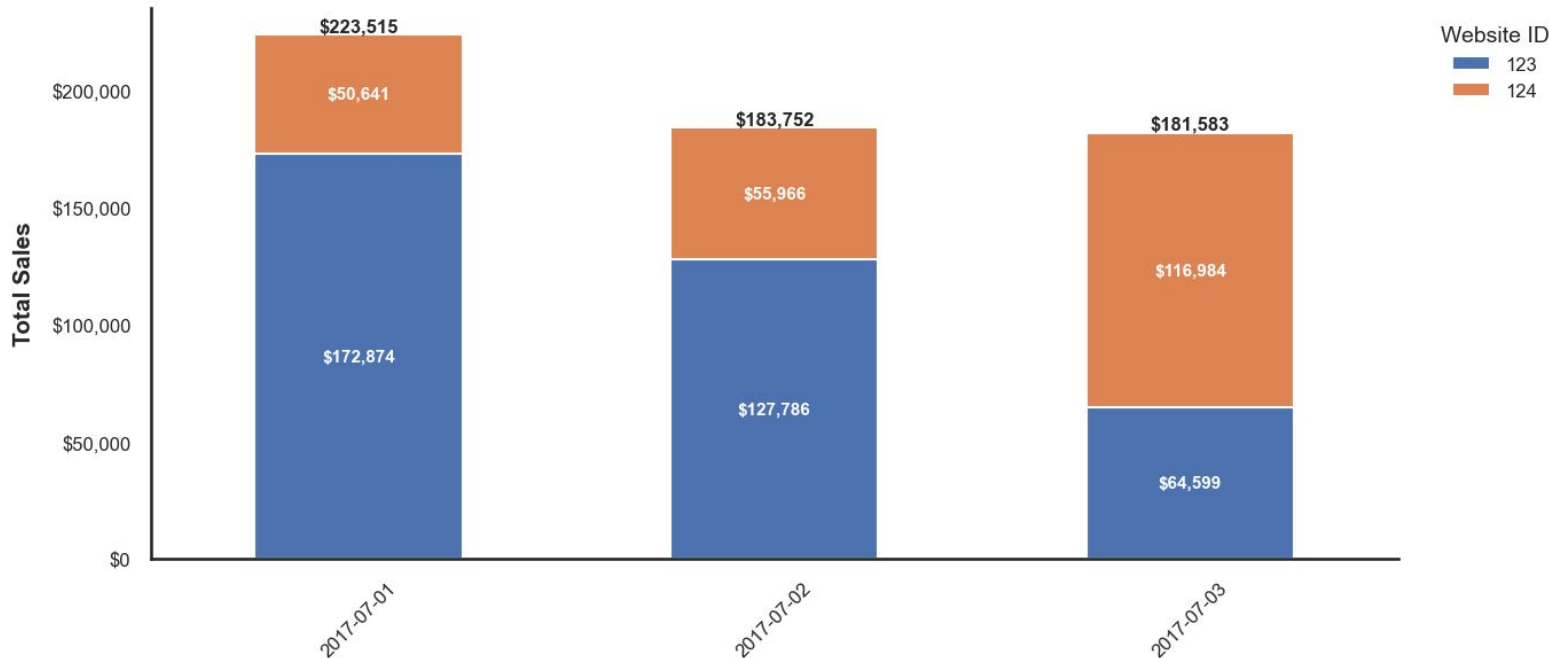
Revisiting the largest transaction on day 3:

- I would expect **1 Bignay to be \$6**
- The checkout amount is \$60,000 **implying 10,000 bignays**
- **Is there an issue when the quantity is > 4 digits?**

THIS ERROR MAY INDICATE A URL FORMATION BUG

4. Are there any interesting purchasing combinations, events, or metrics that are worth reporting and displaying?

Daily Total Sales Composition by Website



Website 124 grows from \$50.6k to \$117.0k, while Website 123 drops from \$172.9k to \$64.6k.

THE TWO WEBSITES ARE NOT MOVING THE SAME WAY

5. Can you predict the total sales number for 2017-07-04? How certain is the predicted number?

With only three days of data, I did not want to force a complicated forecasting model. Instead, I used a **small set of simple baseline estimates** to anchor the prediction.

Method	Forecasted July 4 Sales	Interpretation
Prior-day forecast	\$181,583	Uses July 3 as the best available proxy for July 4.
Recent 2-day average	\$182,668	Uses the average of the two most recent normalized daily totals.
Transaction count $\times$ recent AOV	\$182,692	Combines recent transaction volume with recent average order value.
Sign-error adjustment	\$181,991	Shows the small effect of treating July 3's -\$12 values as sign errors.

I predict July 4 total sales will be approximately **\$182,500**, with an uncertainty range of roughly \$181,500 to \$183,000.

This prediction is low confidence. Even more so because July 4 is a U.S. holiday and customer behavior may not match a normal day.

WITH ONLY 3 DAYS OF DATA, A SIMPLE FORECAST IS THE BETTER CHOICE

6. Is there any additional information, data, or access that would make your prediction better?

Additional data or access	Why it would improve the prediction
More historical transaction data	Needed to perform more sophisticated modeling and estimate normal day-to-day variation, weekday patterns, seasonality, and baseline sales volatility.
Prior holiday / July 4 sales history	Would help estimate whether July 4 tends to increase or decrease online sales for these products.
Confirmed timezone/reporting rules	Avoid date-boundary ambiguity
App/error logs	Validate malformed URLs and large transactions

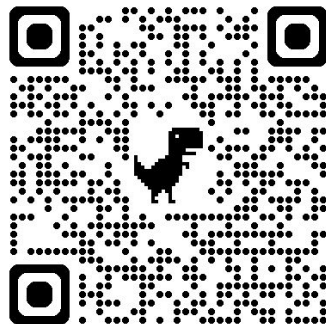
MORE HISTORICAL DATA WOULD SUPPORT A STRONGER FORECAST

Thank You

Questions?

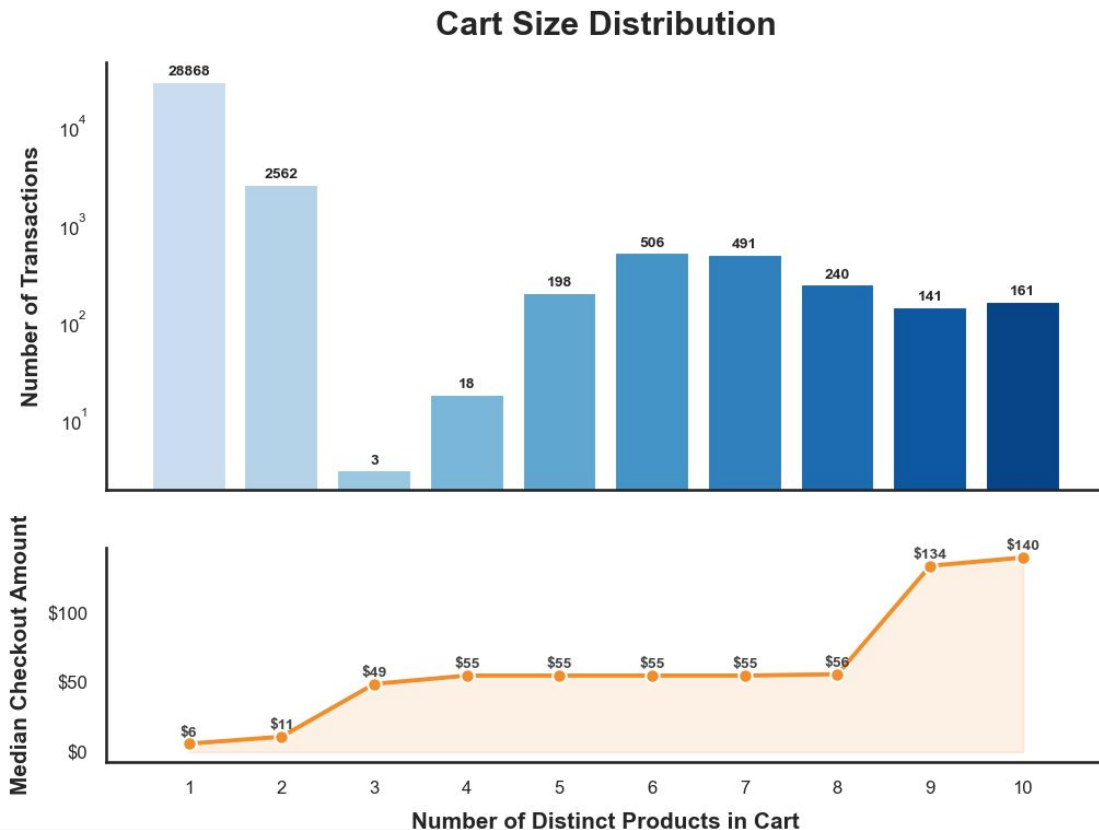
Analysis, code, and supporting materials are at:

<https://github.com/melissadale/postie-data-science-presentation>



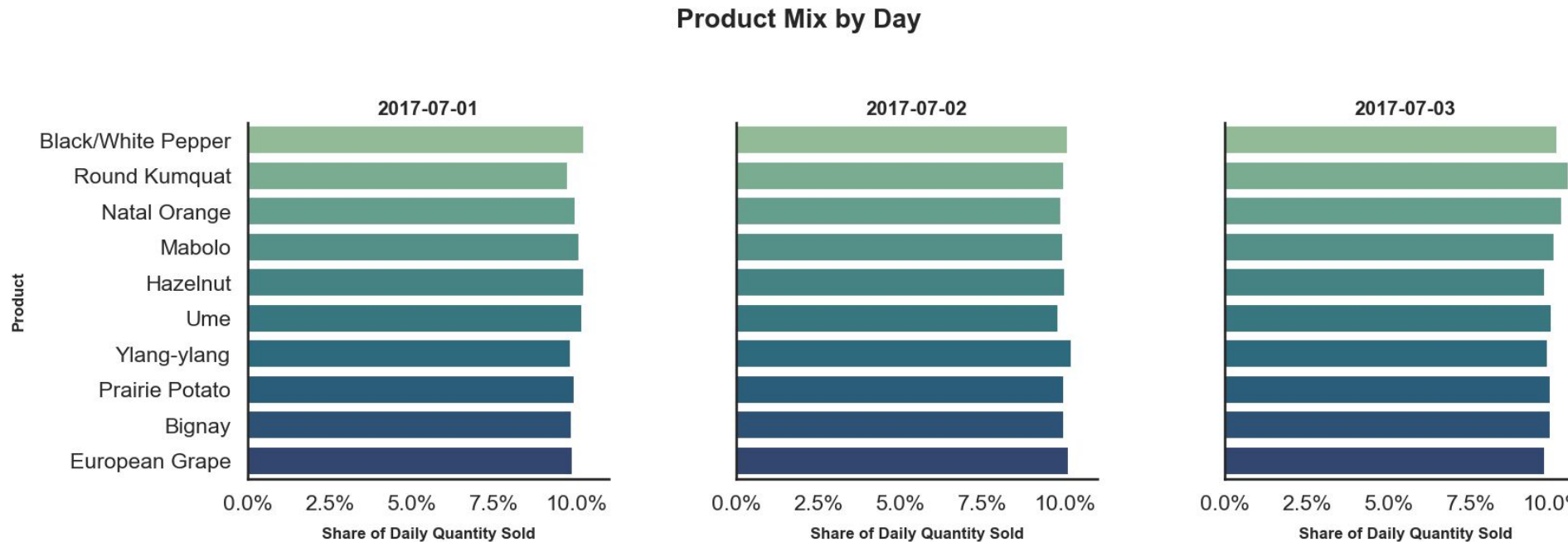
Appendix: What information can be extracted from the URLs? Can product prices be inferred?

Most carts contain only one or two distinct products, while larger carts are less common but have higher median checkout values.



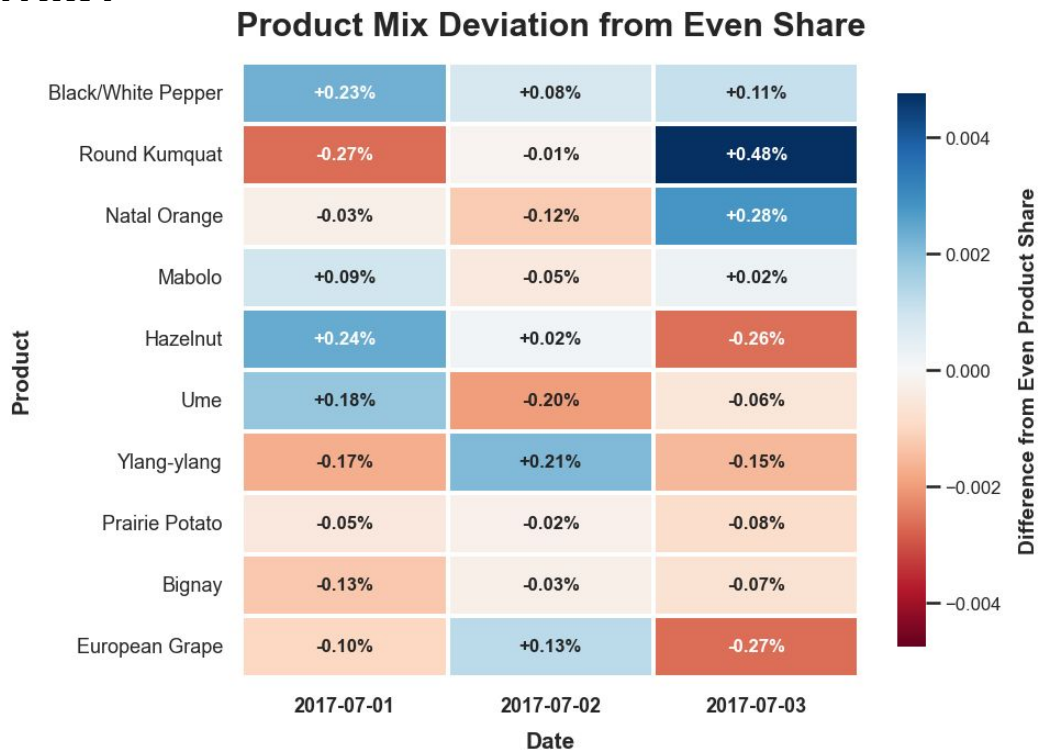
URL-DERIVED CART DATA LETS US ANALYZE HOW PEOPLE ARE BUYING

Appendix: What information can be extracted from the URLs? Can product prices be inferred?



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URL-DERIVED CART DATA LETS US ANALYZE HOW PEOPLE ARE BUYING

Appendix: Can product prices be inferred for negative values?

	Timestamp	Website ID	App Version	Observed Checkout	Observed Absolute Value	Inferred Cart Total	Parsed Cart	Total Quantity	Observed Abs Matches Inferred Total
0	2017-07-03 00:27:19+00:00	124	1.1	\$-12	\$12	\$12	European Grape + Round Kumquat	2.0	True
1	2017-07-03 08:20:25+00:00	124	1.1	\$-12	\$12	\$12	Bignay	2.0	True
2	2017-07-03 01:28:04+00:00	124	1.1	\$-12	\$12	\$12	Ume + Natal Orange	2.0	True
3	2017-07-03 10:01:08+00:00	123	1.1	\$-12	\$12	\$12	Round Kumquat + European Grape	2.0	True
4	2017-07-03 03:10:05+00:00	124	1.1	\$-12	\$12	\$12	Round Kumquat + Black/White Pepper	2.0	True
5	2017-07-03 11:15:33+00:00	124	1.2	\$-12	\$12	\$12	Round Kumquat + Black/White Pepper	2.0	True
6	2017-07-03 11:18:54+00:00	124	1.2	\$-12	\$12	\$12	Round Kumquat + Ylang-ylang	2.0	True
7	2017-07-03 04:37:32+00:00	123	1.2	\$-12	\$12	\$12	Round Kumquat + European Grape	2.0	True
8	2017-07-03 11:44:53+00:00	124	1.2	\$-12	\$12	\$12	Natal Orange	2.0	True
9	2017-07-03 05:10:47+00:00	123	1.2	\$-12	\$12	\$12	Hazelnut + Mabolo	2.0	True
10	2017-07-03 05:21:34+00:00	123	1.2	\$-12	\$12	\$12	Round Kumquat + European Grape	2.0	True
11	2017-07-03 12:33:22+00:00	124	1.2	\$-12	\$12	\$12	Bignay	2.0	True
12	2017-07-03 05:36:29+00:00	124	1.2	\$-12	\$12	\$12	Bignay	2.0	True
13	2017-07-03 05:41:52+00:00	123	1.2	\$-12	\$12	\$12	Round Kumquat + Ylang-ylang	2.0	True
14	2017-07-03 06:30:39+00:00	124	1.2	\$-12	\$12	\$12	Round Kumquat + Black/White Pepper	2.0	True
15	2017-07-03 07:49:50+00:00	124	1.2	\$-12	\$12	\$12	European Grape + Round Kumquat	2.0	True
16	2017-07-03 14:57:28+00:00	123	1.2	\$-12	\$12	\$12	Round Kumquat + Black/White Pepper	2.0	True

INFERRED CART COSTS